Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture : Convergence of Collatz (3n+1) Sequence to the Trivial Cycle A DEFINITIVE PROOF

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The entire dynamics of the Collatz (3n+1) system can be exactly represented by a dynamically-evolving graded-algebraic-structure of an ideal-based-filtration-scheme; with divisibility-classes for modulo-3 quotient-semiring generated by the primitive root 2; along with a filtration-shifting-global-affine-transformation, f(x)=(3x+1) when x is a positive odd number; achieving a Euclidean expansion shift to the topmost coprime-layer of that filtration-scheme; while also avoiding the modulo-multiple-layer (zero-layer) and all the nilpotent-layers with nilpotent-elements (dead-end zero-divisors) like (6m-3) and (6m-0). The trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ is bypassed by an initialization-phase for this filtration-scheme; starting with directly the modul-9 coprime-layer. This design for the system-dynamics-framework by itself provides a definitive-proof of the Collatz conjecture, establishing that the entire map of the Collatz-system-dynamics is exactly represented by this dynamically-evolving graded-algebraic-structure defined on the set-of-all-positive-integers; neither missing any natural-number nor including any extraneous-elements; and therefore, providing both the necessary-&-sufficient condition simultaneously in one single sweep.

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